System Blueprint (*a.k.a.* "Design Doc")

TNPG: SARDines

project: NerdyMap

Target ship date: {2026-06-01}

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roster:

Name	Email	Primary Role	Secondary Role
Rohan Sen	rohans33@nycstudents.net	PM	Backend
David Lee	davidl542@nycstudents.net	F Student (Innovator)	Trailblazer
Sean Zheng	seanz22@nycstudents.net	F Student (Innovator)	Trailblazer
Araf Hoque	aoanulh2789@nycstudents.net	D Student	Ladies' Man

Summary

A one-stop shop for frequent nyc commuters, or in other words, a multi-purpose transit system guide targetting city folks.

Problem Being Solved

Simple transit maps provided by apps such as Google or Apple Maps provide limited functionality and are often delayed in updating and timing. We are solving this limited functionality through incorporating visualization of trains' live positions, more interactivity, and more specific details.

Target Users

Who will use this system? - People who utilize public transportation - People who like cool maps

Why This Project Matters

Many transit applications such as those integrated into Google/Apple Maps show limited information about the subway system, and isn't immediately intuitive. With NerdyMap, users can see valuable information like live train positions, delays, and station-specific information right at a glance.

Minimum Viable Product (MVP) Scope

Core Features (Required for Final Submission)

Features that **must** be completed: 1. An interactive map showcasing transit routes in a given area, with realtime train positions mapped (NYC) 2. List of active delays across the city 3. Wayfinding between different stops

Stretch Features (Only if MVP is Complete)

1. Add LIRR Support
2. Customizeable Interface (Pin lines, change UI parameters, etc)
3. Map changes based on delays

Explicit Non-Goals

We don't want to recreate Google/Apple Maps. Our intent is to develop a more in-depth transit map, rather than just a clone. We'll have real-time delay information pinned to the side of the screen, a live map of subway trains moving around the city, and more detailed stats for each station/train (crowding, speed, elevator status, etc).

Features intentionally excluded: - Multiple cities being integrated, since we want to focus on NYC's needs specifically and its unique system

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Technology Stack

Layer	Selected Tool
Backend Framework	Flask
Frontend Framework	bootstrap
Database	SQLite
Authentication	Flask sessions
ORM / DB Library	SQLAlchemy

Why This Stack Was Chosen

We chose this stack because it matches our team's experience and project needs. Flask is powerful enough to handle routes, API requests, sessions, and transit data without being too complex. Bootstrap helps us quickly build a clean, responsive interface using pre-styled components. SQLite works well because we have more experience with it than MongoDB, and it can store data like users, saved stops, pinned routes, and preferences. Flask sessions support basic login and user-specific features, while SQLAlchemy helps us connect our Python code to the database in a cleaner and more organized way. Overall, this stack lets us focus on building NerdyMap's transit features instead of spending too much time learning unfamiliar tools.

Team Ownership Plan

Each member must own meaningful deliverables.

Team Member	Primary Ownership	Secondary Ownership	Specific Deliverables
Rohan Sen	Frontend (transit map anims)	Backend (transit API integration)	Get the transit map of NYC up and running, toggle options
David Lee	Backend (transit API integration)	Frontend (UI/UX)	Get data pipeline up and running, backend to frontend comms
Sean Chen	Backend (any other API integrations)	Frontend (UI/UX)	Along with David, get API integrations running
Araf Hoque	Frontend (transit map)	Frontend (UI/UX)	Help Rohan on transit map animations (has prior exp from personal passion project), help on frontend too

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Component map

Site map

Mockup

Key User Stories

eg0

As a nerd, I want to watch the trains move around the map so that I can see the city moving around in real-time.

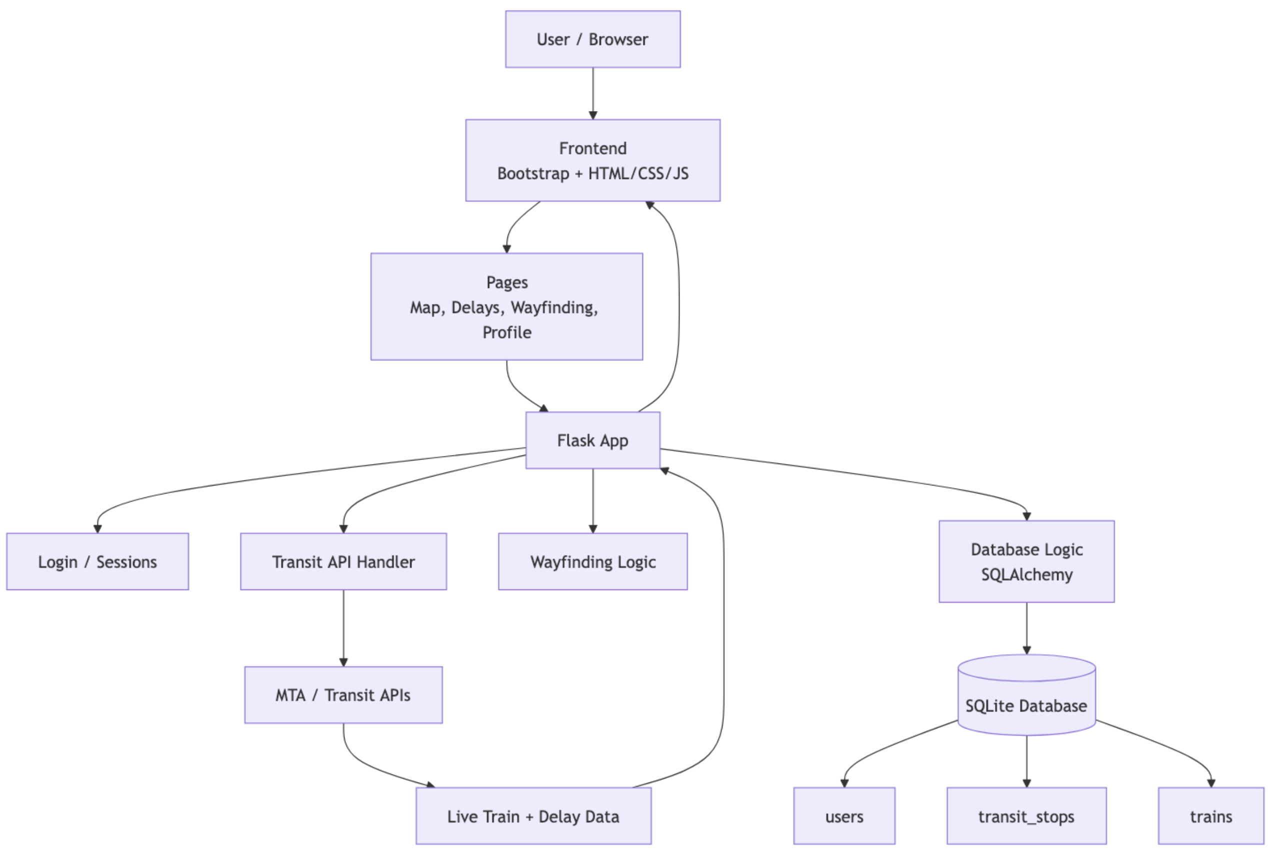

Figure 1: Component map

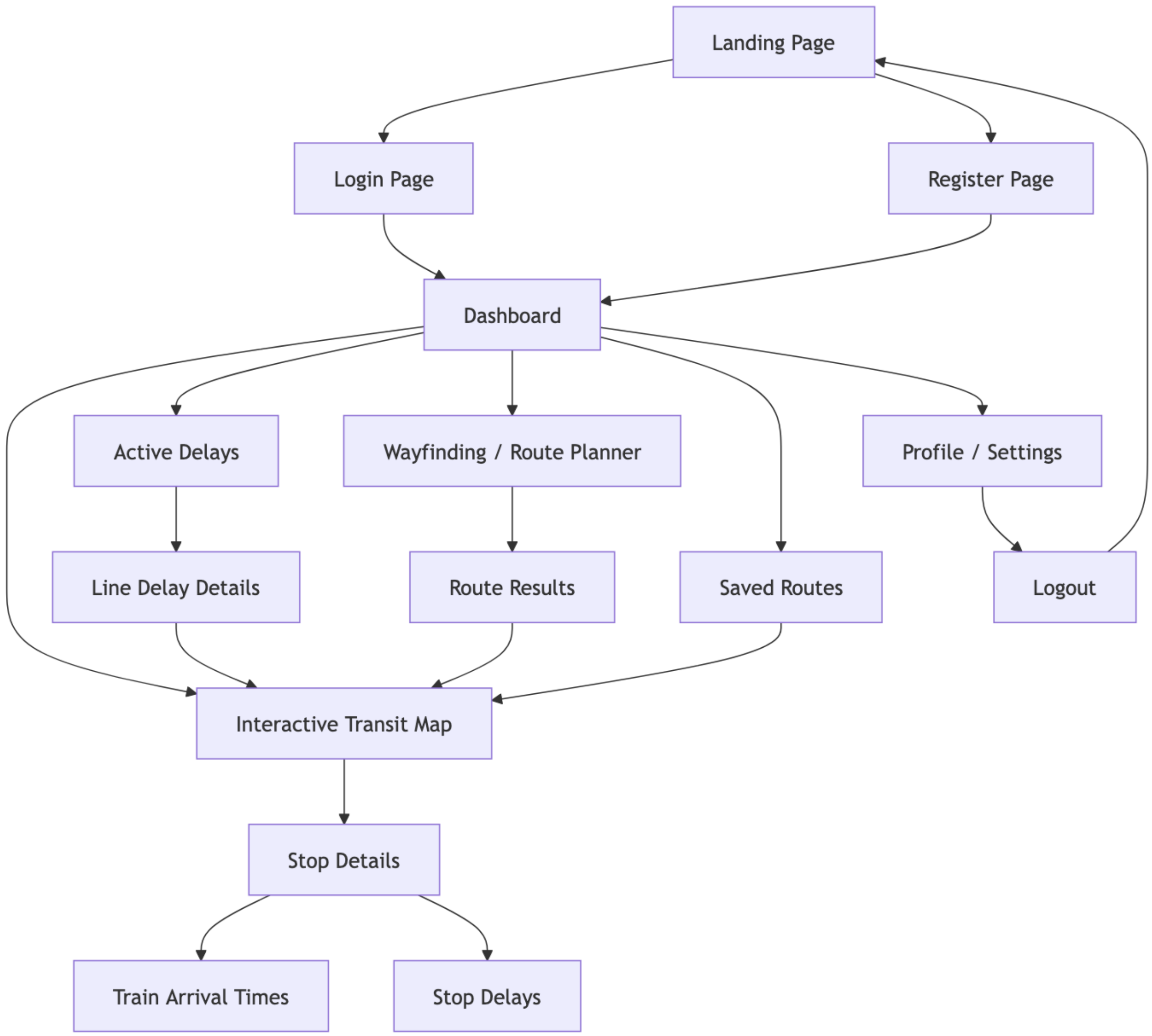

Figure 2: Component map

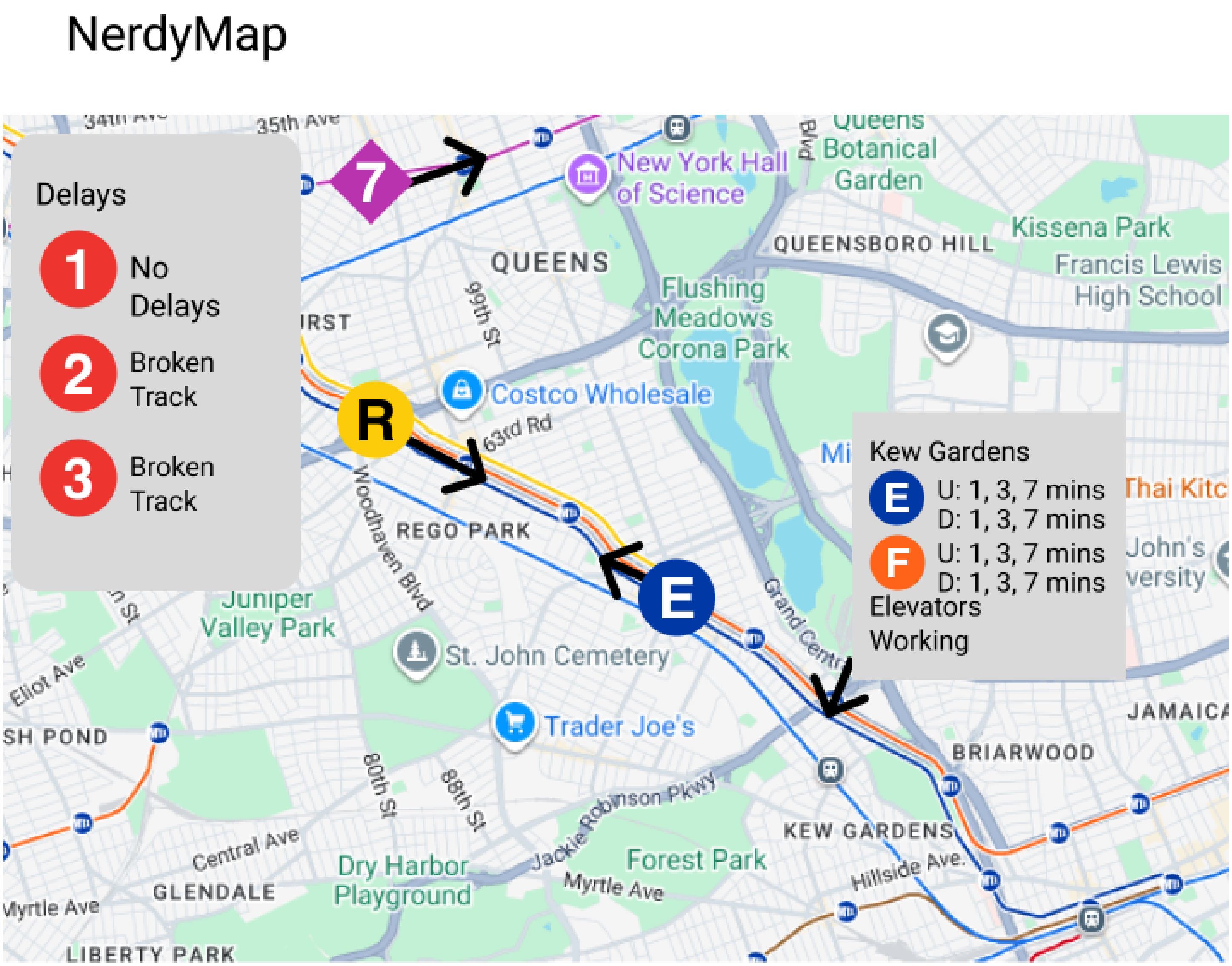
NerdyMap
Delays
1 No Delays
2 Broken Track
3 Broken Track
7
R
E
Kew Gardens
E U: 1, 3, 7 mins
D: 1, 3, 7 mins
F U: 1, 3, 7 mins
D: 1, 3, 7 mins
Elevators Working
QUEENS
REGO PARK
New York Hall of Science
Queens Botanical Garden
QUEENSBORO HILL
Kissena Park
Francis Lewis High School
Flushing Meadows Corona Park
Costco Wholesale
63rd Rd
99th St
Woodhaven Blvd
Juniper Valley Park
St. John Cemetery
Trader Joe's
Eliot Ave
80th St
88th St
Central Ave
GLENDALE
Dry Harbor Playground
Jackie Robinson Pkwy
Forest Park
Myrtle Ave
KEW GARDENS
Hillside Ave.
BRIARWOOD
LIBERTY PARK
35th Ave

Figure 3: UI

eg1

As a New Yorker, I want to see where my train is so that I know how long I have to wait in the station, since the time estimates aren't as accurate.

eg2

As a parent, I would want to track a specific train so that I can watch my kids come home and prepare for them to do so.

APIs

MTA

https://api.mta.info/#/subwayRealTimeFeeds
https://api.mta.info/#/serviceAlerts
https://api.mta.info/#/EAndEFeeds https://api.mta.info/#/serviceAlerts

Google Maps

https://developers.google.com/maps/documentation/javascript/overview
https://developers.google.com/maps/documentation/routes/transit-route.

Database Design

users

DATA_TYPE	VAR_NAME	KEY_STATUS	NOTES
TEXT	username	PK	unique identifier of user
TEXT	password		
DATE	creation_date		
TEXT	saved_routes		routes user's saved/favorited

transit_stops

DATA_TYPE	VAR_NAME	KEY_STATUS	NOTES
TEXT	stop_name	PK	unique identifier of transit stop
TEXT	possible_trains		all possible trains that can stop here
TEXT	stopped_trains		status of trains currently stopped here
TEXT	directions		direction of train stopped (uptown/downtown), corresponds w/ stopped_trains
TEXT	delays		any delays relayed by API calls

trains

DATA_TYPE	VAR_NAME	KEY_STATUS	NOTES
TEXT	train		what train line it is
INT	train_id		considering dupes of trains
TEXT	arrival_times		arrival times for this specific train

DATA_TYPE	VAR_NAME	KEY_STATUS	NOTES
TEXT	departure_times		departure times for this specific train
TEXT	delays		delays for this line (overall) or train in specific (running behind schedule)

Testing Plan

- Have members from other teams test our transit map animations, get feedback for client-side before near-final deadine so we can tweak and adjust
- Rigorously test UI/UX for different screens and dimensions so the animations are not buggy on diff. devices (especially transit map); important to make UI/UX not stagnant on the singular windows dim. we're working on
- Ensure API calls on both dev and user ends work so it doesn't magically break during demos, etc. by testing multiple times before, during, and after changes to codebase (A/B testing)

Timeline

Week 1 Goals: Basic map functionality, pull data from the MTA and other APIs.

Week 2 Goals: Plotting trains on the map, rendering their movement, use websockets.

Week 3 Goals: UI perfection, debugging, hopefully no major features pending.

Internal Deadlines:

- Determine all the APIs we need (5/15/26 –> 5/15/2026)
- Have API keys, test calls set up (5/18/26 –> 5/15/26)
- Basic map functionality up on frontend (5/22/26 –> 5/15/26)
- More complex and detailed functionalities/tools up on frontend (5/25/26 –> PLANNED: 5/22/26)
- Catch-up week on any functionalities that need more work or incomplete (5/29/26 –> PLANNED: 5/22/26)
- Work on UI/UX to make it intuitive and easy to work with (5/29/26 –> PLANNED: 5/25/26)
- Expand to other cities (BIG IF)

Completion Criteria (*a.k.a.* "Definition of 'Done'")

Project is considered complete when all of the following are true: 1. Transit map of NYC is fully interactable and viewable of transit disruption panel, destination tracking/steps 2. API calls work 24/7 whenever service is called upon to use, doesn't take too long to load each call 3. Frontend UI/UX is intuitive and easy to work with for users, not just the devos who made the service.

Open Questions

- Will we try to implement multiple cities? Presumably, if we can implement NYC transit tracking, we could for other cities, too.

Appendix

None at the moment.

Other

None at the moment.